



جامعة نيويورك أبوظبي
NYU ABU DHABI

Advanced Circuits

ENGR-UH 2311 | Spring I - 2025

Experiment: Lab 1

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Lab Report: DC Operating Points Analysis of Circuits in Cadence**Objectives**

The primary objectives of this lab are to utilize the Cadence environment to create and analyse circuit schematics, define the parameters of electrical components, set up and configure the simulation environment, run simulations to examine circuit behaviours, annotate and identify the DC operating points of circuits, plot various variables and their properties, and export high-resolution images of the schematics and plots.

Materials and Equipment

1. Cadence Software

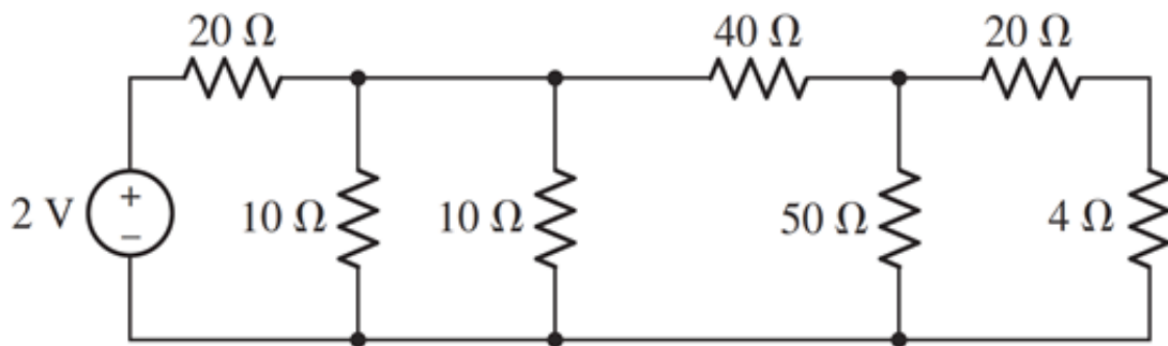
Diagram

Figure 1: Circuit 1: Schematic with DC Voltage Source

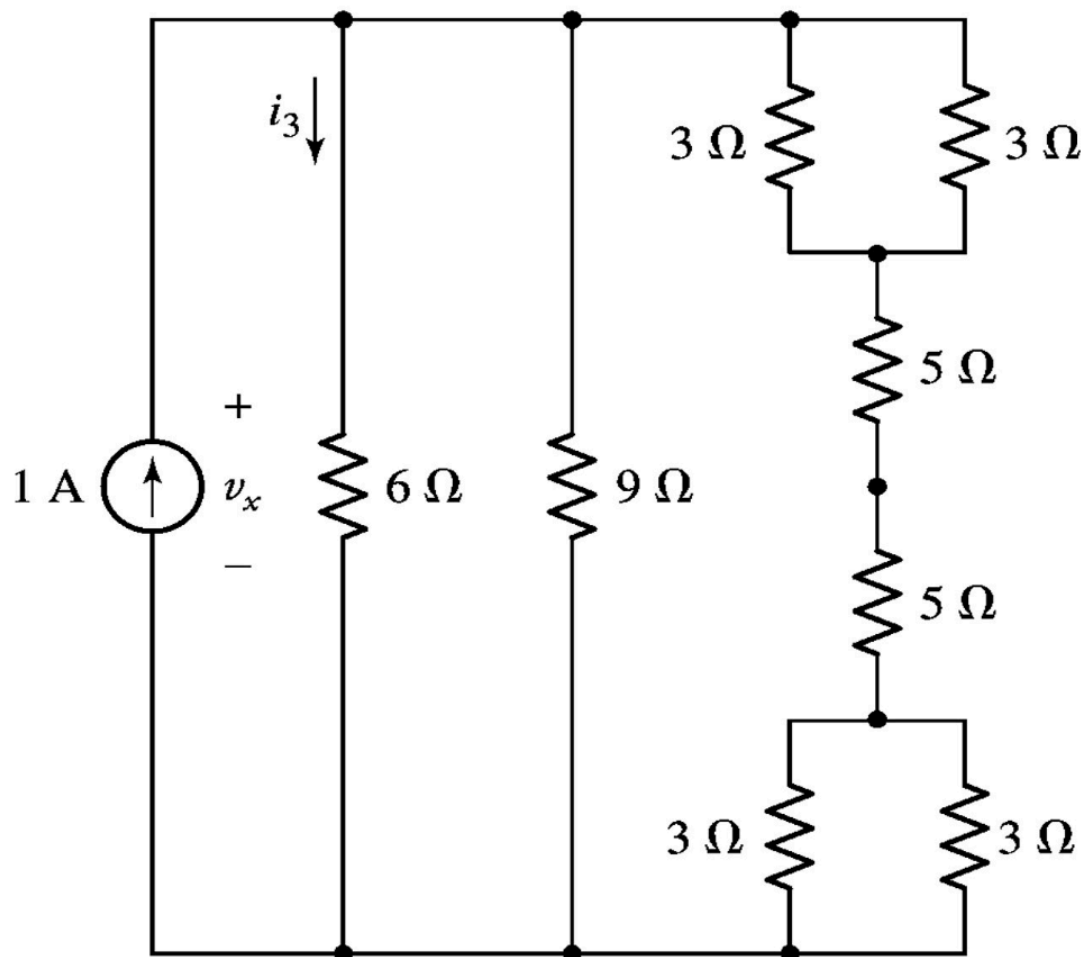
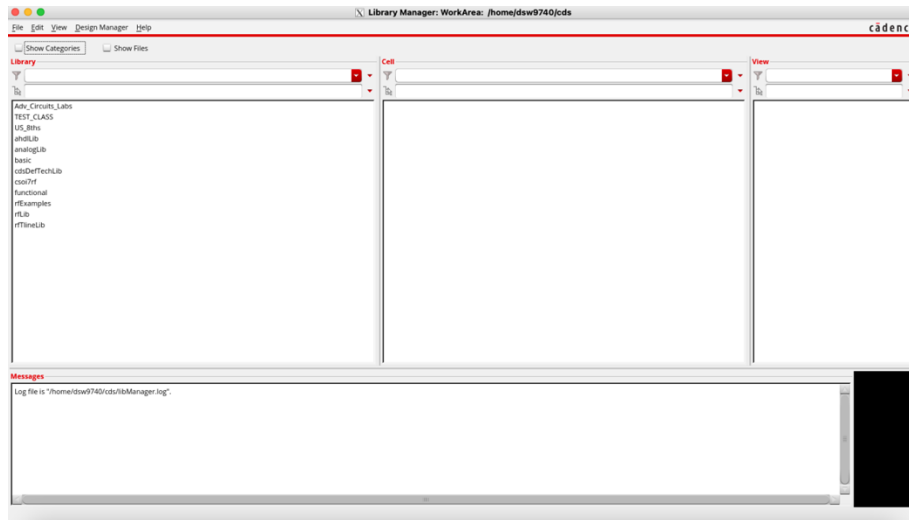


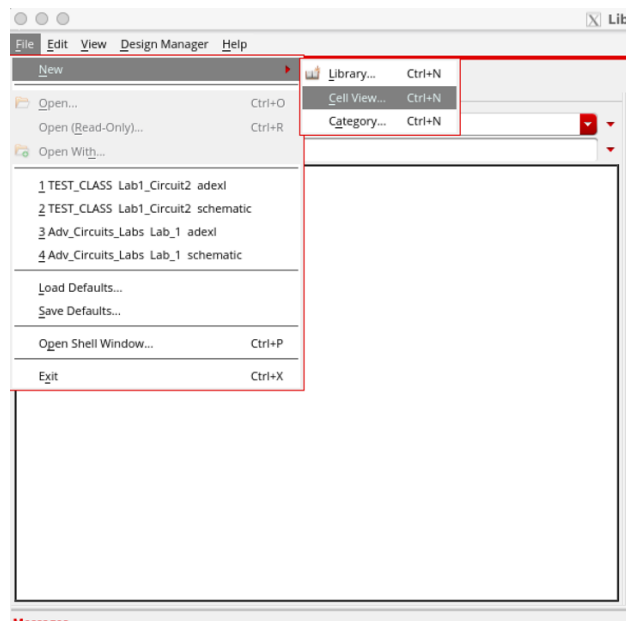
Figure 2: Circuit 2 Schematic with DC Current Source

Procedure

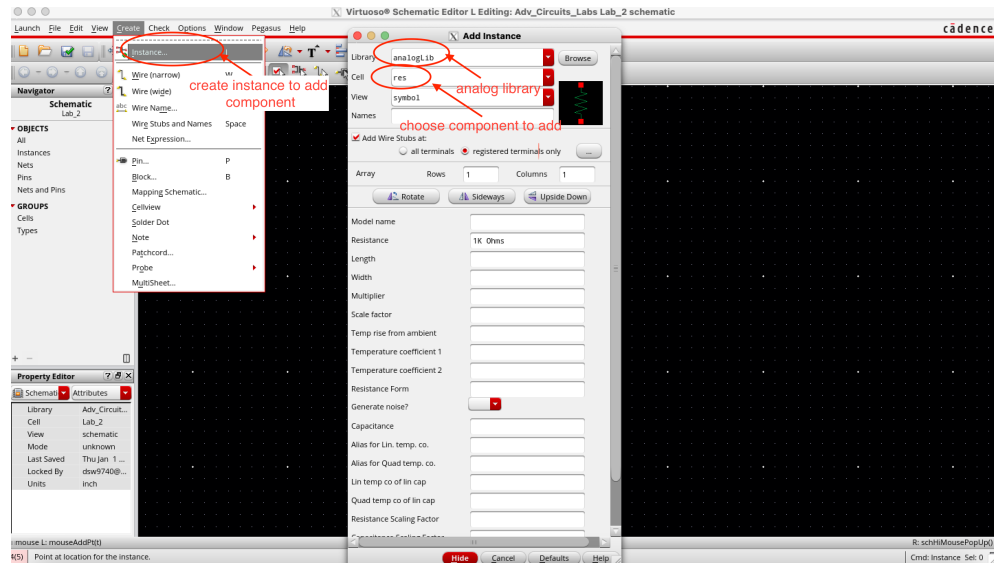
1. The Cadence Virtuoso platform was launched from the server.



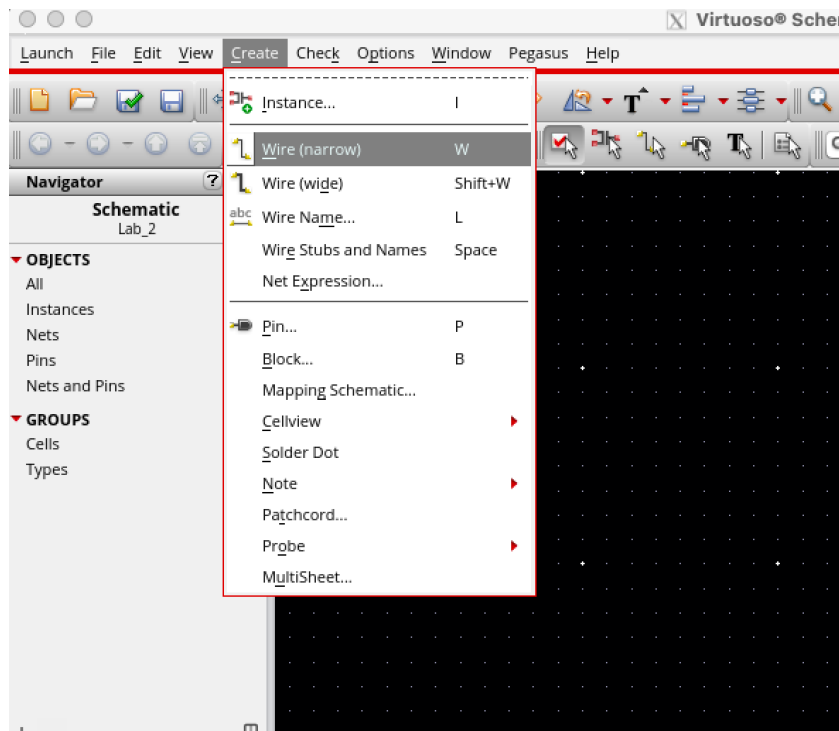
2. Within the TEST_CLASS library, a new cell was created and labelled.



3. In the newly opened Schematic Editor, components were added, by navigating to **Create > Instance** and selecting the **analogLib** library.



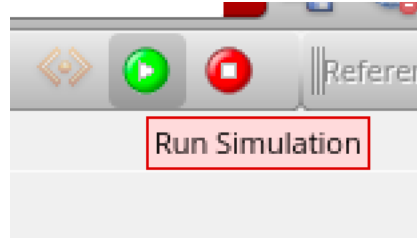
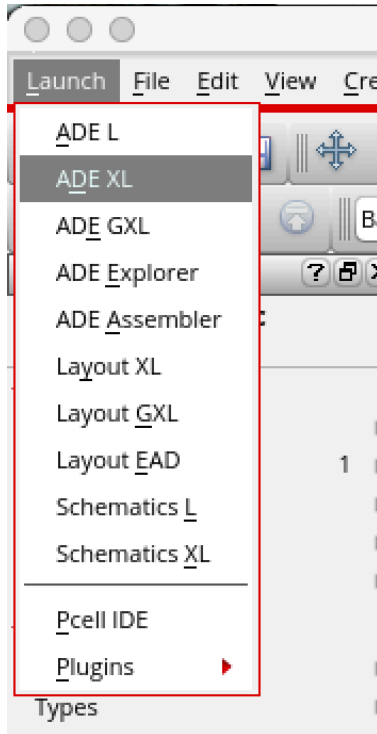
4. Voltage sources (vdc), resistors (res), current sources (idc) were added following the Schematics in Figure 1 and the parameters were adjusted as required.
5. Wires were added by pressing **W** or going to **Create > Wire (narrow)** to complete the circuit connections.



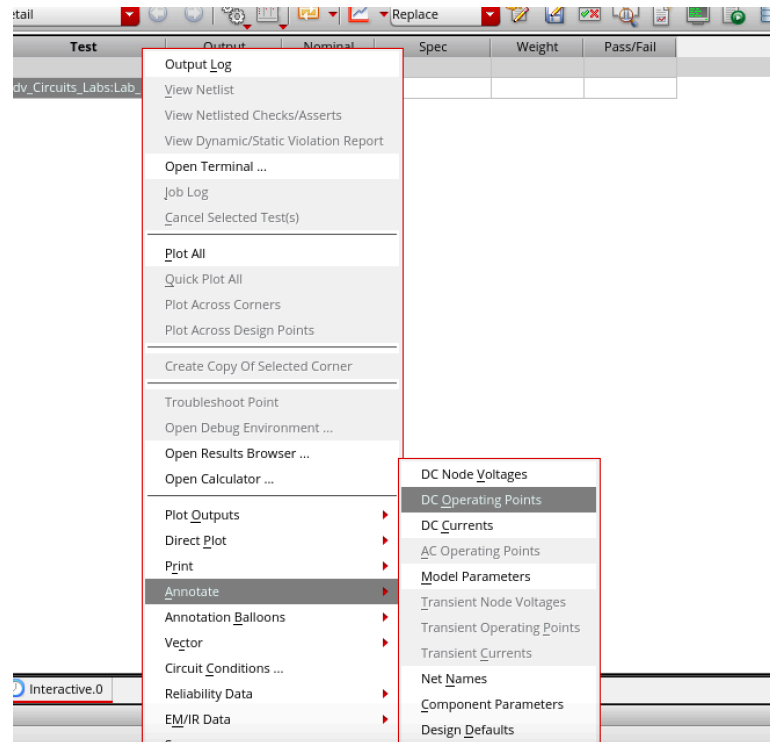
6. Errors were checked for by clicking **Check and Save** found in the upper left side of the ribbon and any connection errors were resolved.



7. A simulation was set up in ADE XL by navigating to **Launch > ADE XL** and configuring the setting to run a dc analysis. The simulation was then set to run by clicking the green button in the ribbon.



8. After the simulation was complete, the circuit was annotated by navigating **Annotate > DC Operating Points**.



9. The steps were repeated for circuit schematic 2 in Figure 2.

Simulation Results

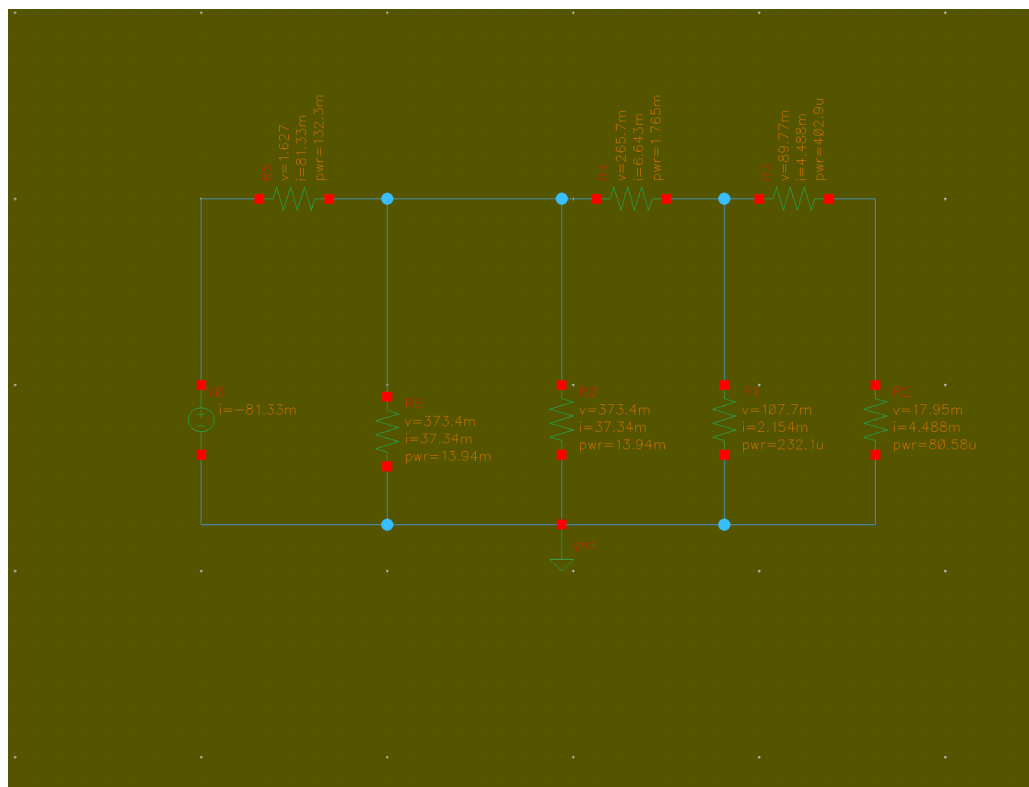
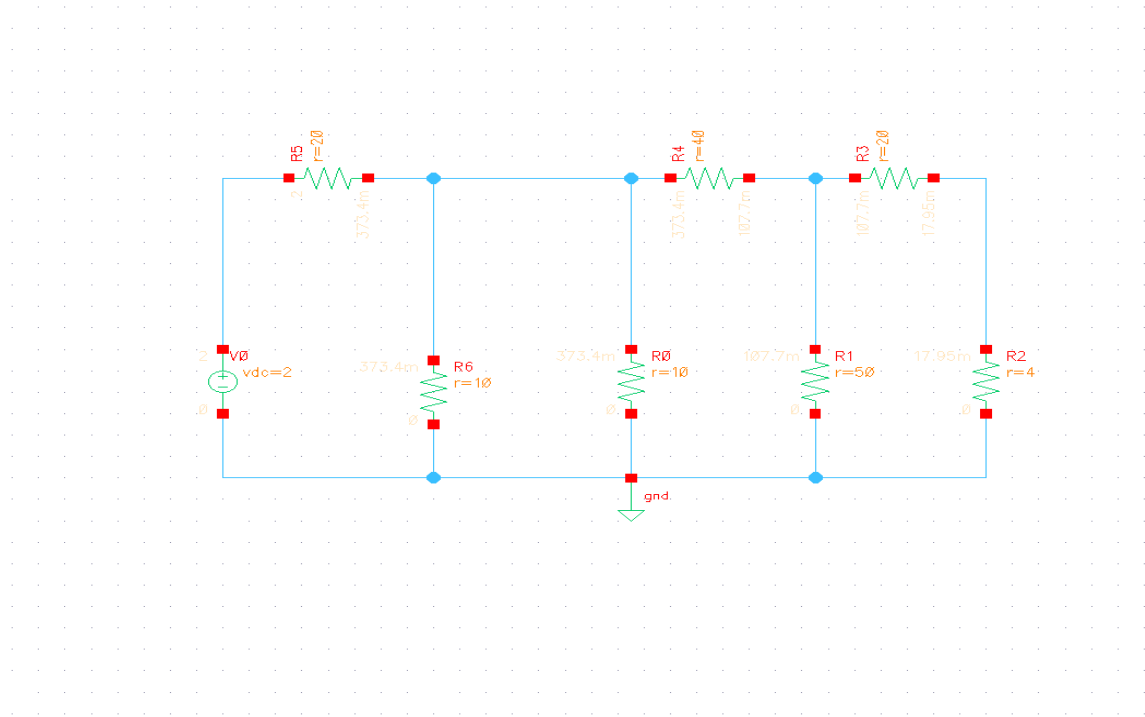


Figure 3: Simulation Results For Circuit 1 Schematic

Resistor .No.	Resistance(R/Ohms)	Voltage (V/mV)	Current (I/mA)
R0	10	373.4	37.34
R1	50	107.7	2.15
R2	4	17.95	4.49
R3	20	89.77	4.49
R4	40	265.7	6.64
R5	20	1.63	81.3
R6	10	373.4	37.3

Table 1: DC Operating Points Obtained from Simulation For Circuit 2 Schematic

From Figure 3, the power dissipated by 4 Ohm Resistor (R2) was found to be 80.58 μ W.

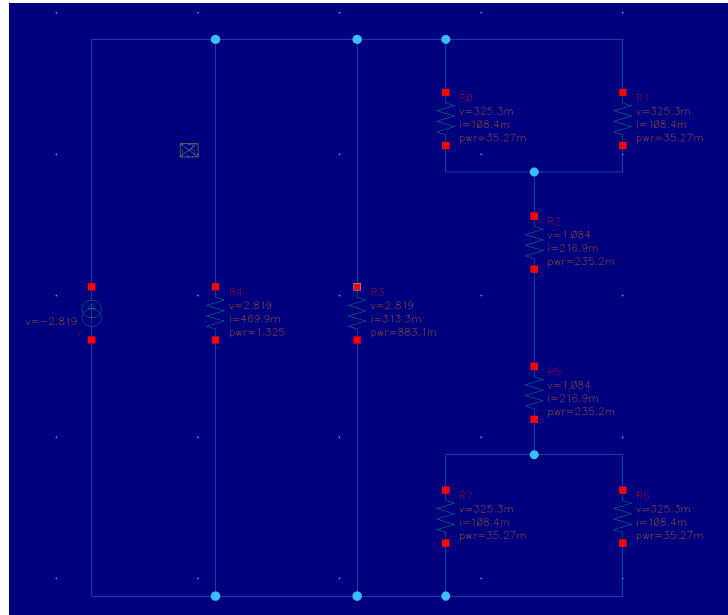


Figure 4: Simulation Results For Circuit 2 Schematic

Resistor .No.	Resistance(R/Ohms)	Voltage (V/mV)	Current (I/mA)
R0	3	325.3	108.4
R1	3	325.3	108.4
R2	5	1084	216.9
R3	9	2819	313.3
R4	6	2819	469.9
R5	5	1084	216.9
R6	3	325.3	108.4
R7	3	325.3	108.4

Table 2: DC Operating Points Obtained from Simulation For Circuit 2 Schematic

Conclusion

The Cadence environment was used to simulate the DC Operating Points for the given circuit schematics.